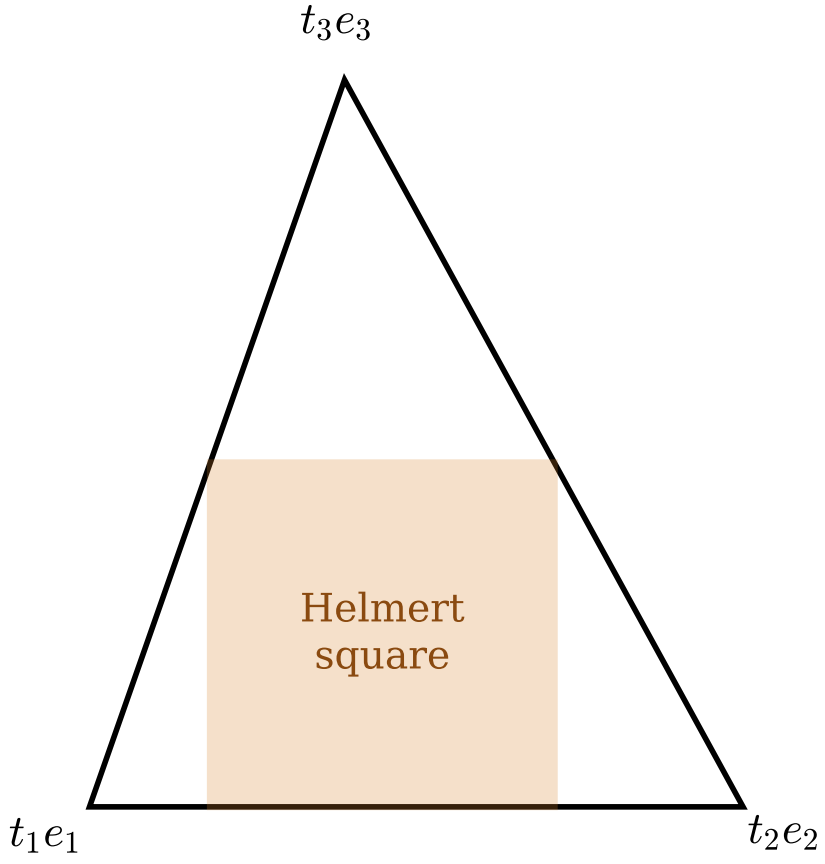
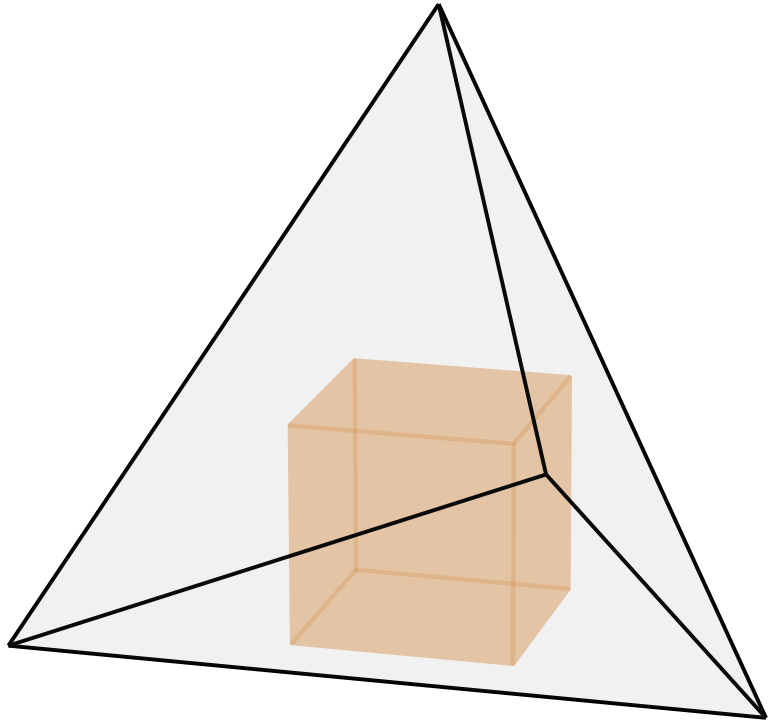
$d = 3$: a square in an acute triangle




$d = 4$: Croft's cube